AI ENGINEERING · KNOWLEDGE SHARE

Five Problems, One Deep Dive

What actually goes wrong when we pull structured data out of PDFs — stated plainly — and then one of them taken all the way down: why rows go missing, how to detect it, and the math for estimating what we never saw.



THE MAP

The Five, Grouped

INPUT

1. The model can only read what we handed it

2. Tables don't survive the trip

SPEC

3. We never agreed what the field means

TRUST

4. Rows go missing and nothing tells us

5. Wrong answers look exactly like right answers

Two are input problems, one is a specification problem, two are trust problems. Only the last two are invisible without instrumentation.

PROBLEM 1 · INPUT

The Model Only Sees What OCR Gave It

ON THE PAGE

Tier	Volume	Rate
1	0 – 2,000	2%
2	2,001 – 5,000	4%
3	5,001 +	6%

→

WHAT WE PASS TO THE PROMPT

```
Tier Volume Rate 1 0 2,000 2% 2 2,001 5,000
4% 3 5,001 + 6%
```

Row boundaries gone. Column membership gone. **Is 4% tier 2's rate, or tier 3's?**

Test it yourself: print the exact string we send, and try to rebuild the table by eye. If you can't, no prompt can.

PROBLEM 2 · INPUT

Tables Are Held Together by Whitespace

RELATIVE DIFFICULTY (illustrative)

Borderless



Column edges exist only in the eye.

Merged cells



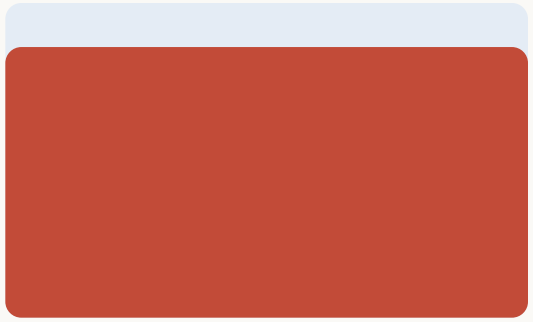
One label owning three rows.

Nested headers



Two header rows, one meaning.

Page-spanning



Header on page 3, rows on page 4.

Nothing in the file marks a cell boundary. Recovering rows and columns is a separate job from reading the values — and it has to happen first.

PROBLEM 3 · SPEC

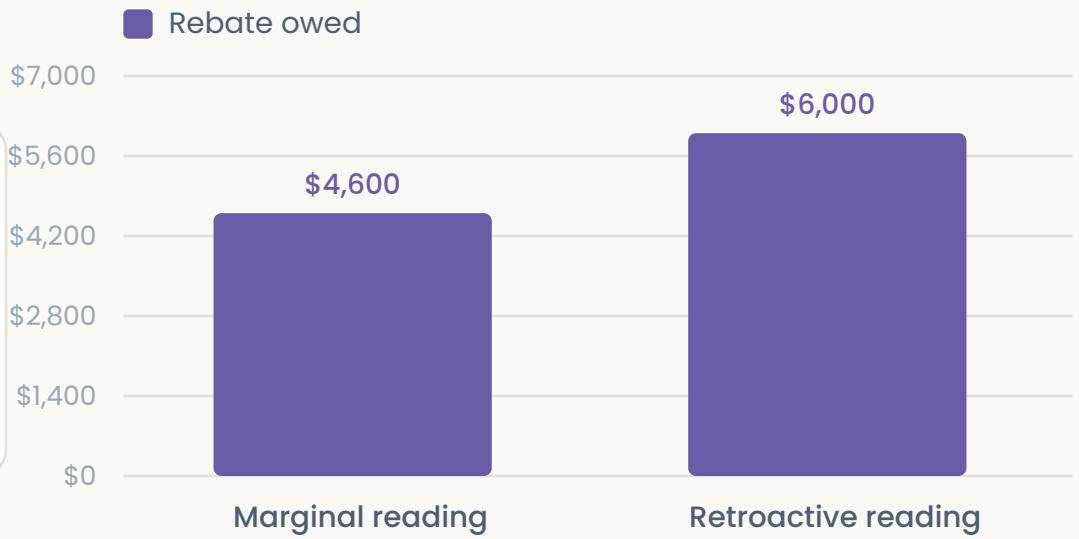
Same Table, Two Answers

10,000 units at $10, tiers 2% / 4% / 6%. Is the top rate marginal or retroactive? The document rarely says.

Marginal 2,000×$10×2% + 3,000×$10×4% + 5,000×$10×6% = **$4,600**

Retroactive $100,000 × 6% = **$6,000**

30% difference on one undefined word.



If we haven't decided, the model decides — and not always the same way twice.

PROBLEM 4 · TRUST

Rows Go Missing, Silently

DOCUMENT · 6 TIERS

- Tier 1
- Tier 2
- Tier 3
- Tier 4
- Tier 5
- Tier 6

JSON · 5 TIERS

- Tier 1
- Tier 2
- Tier 3
- missing
- Tier 5
- Tier 6

THE SCREEN

- ✓ HTTP 200
- ✓ Schema valid
- ✓ 5 tiers rendered
- ✓ No warning shown
- ✗ **The data is wrong**

A wrong value eventually gets caught by a human who knows the account. A missing row has no advocate — there is nothing there to notice.

PROBLEM 5 · TRUST

Failure Doesn't Look Like Failure

A parser that fails

- Returns nothing
- Throws an exception
- Someone gets paged
- **Loud, obvious, fixed**

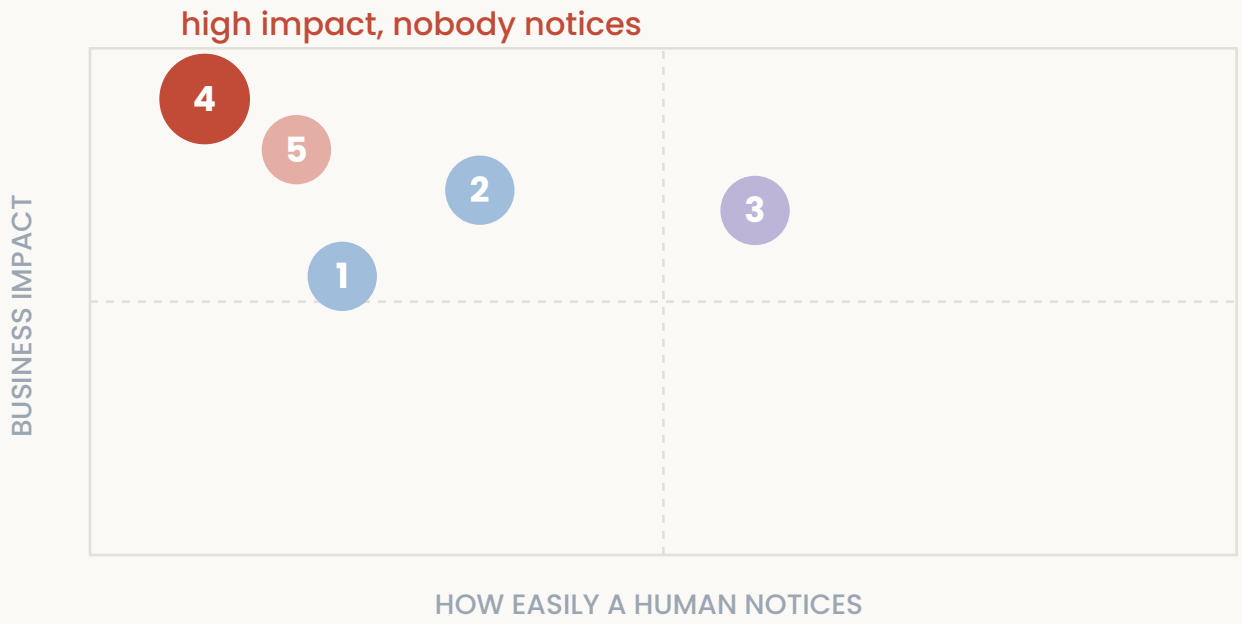
A model that fails

- Returns a number
- Correct JSON shape
- Passes validation
- **Quiet, plausible, shipped**

We cannot tell a right answer from a wrong one by looking at the output. That is the whole reason confidence and completeness need their own machinery.

THE CHOICE

Why We're Going Deep on #4



Completeness, because:

- Nobody has thought about it
- Demonstrable live in 30 seconds
- Fixable without model changes
- Has real math behind it
- Externally validated this year

Problem 5 is the richer topic, but it needs labelled data we don't have yet. Completeness is the gateway.

DEEP DIVE · MECHANISM

Three Ways a Row Disappears

Page-break context loss

The table runs onto page 4. Those rows lost their header, so the model drops or misassigns them.

FIX Per-page extraction with forced header carry-forward

Chunk boundary destruction

A naive chunker splits a record. Each half is processed alone; neither sees the whole row.

FIX Chunk on document structure, not token count

Generation truncation

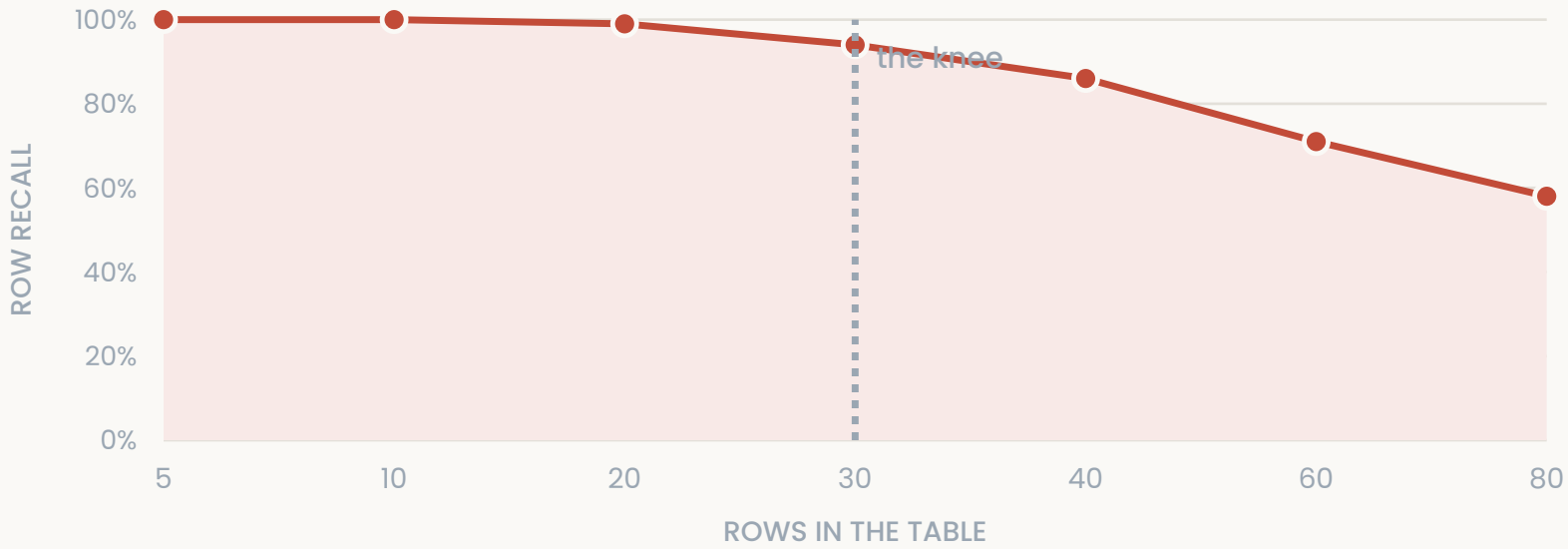
The model stops early while emitting a long JSON array — not a context limit, a degradation.

FIX Split the list; extract in bounded batches

Three mechanisms, three different fixes — diagnosing which is most of the work.

DEEP DIVE · EVIDENCE

Recall Falls Off a Cliff



Shape, not numbers.

Recall is near-perfect on short tables, so every demo passes. It degrades on the long ones, which is where the money is.

Illustrative — replace with our own measurement.

Externally corroborated: recent benchmark work finds commercial vision models do well on short documents but often truncate record lists on long ones.

LAYER 1 · DETECT

Structural Invariants Catch It Free



Tier contiguity

No gaps or overlaps between bands

Control totals

“6 tiers”, “page 3 of 7”, stated sums

Region coverage

Every detected table region yields rows

Edge adjacency

Last row near a page bottom flag

LAYER 2 · MEASURE

Synthetic Tables Give Free Ground Truth

1 Generate

Pick tier values
in code

→

2 Render

Emit a PDF,
plus a scanned copy

→

3 Extract

Run our real
pipeline on it

→

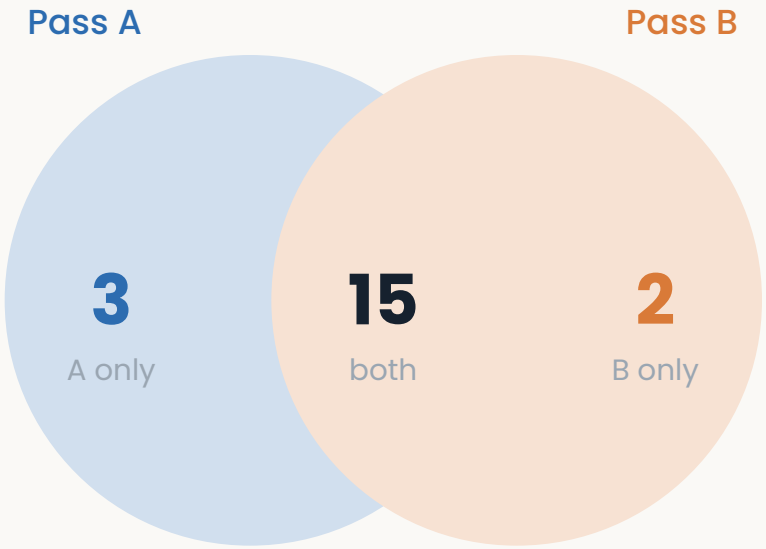
4 Compare

Against values we
started with

Because the values exist **before** the document does, completeness is scorable at any row count with zero labelling cost. This is how you get the curve on the previous slide by Friday.

LAYER 3 · ESTIMATE

Capture–Recapture: Counting Fish



Run two **independent** passes over the same table.

$N = (n_A \times n_B) / m$

$N = (18 \times 17) / 15 = 20.4$

Between them the passes found **20** distinct rows. The estimate says about **20–21** exist.

≈1 row neither pass saw.

The intuition: if two passes disagree a lot, they are each missing a lot — so something is probably missing from both.

THE MATH

Worked, End to End

n_A	rows found by pass A	**18**
n_B	rows found by pass B	**17**
m	rows found by both	**15**
observed	distinct rows seen	**20**

Lincoln–Petersen

N = (18 × 17) / 15 = 306 / 15 = **20.4**

Chapman correction

better behaved on small samples

N = (19 × 18)/(16) - 1 = 21.4 - 1 = **20.4**

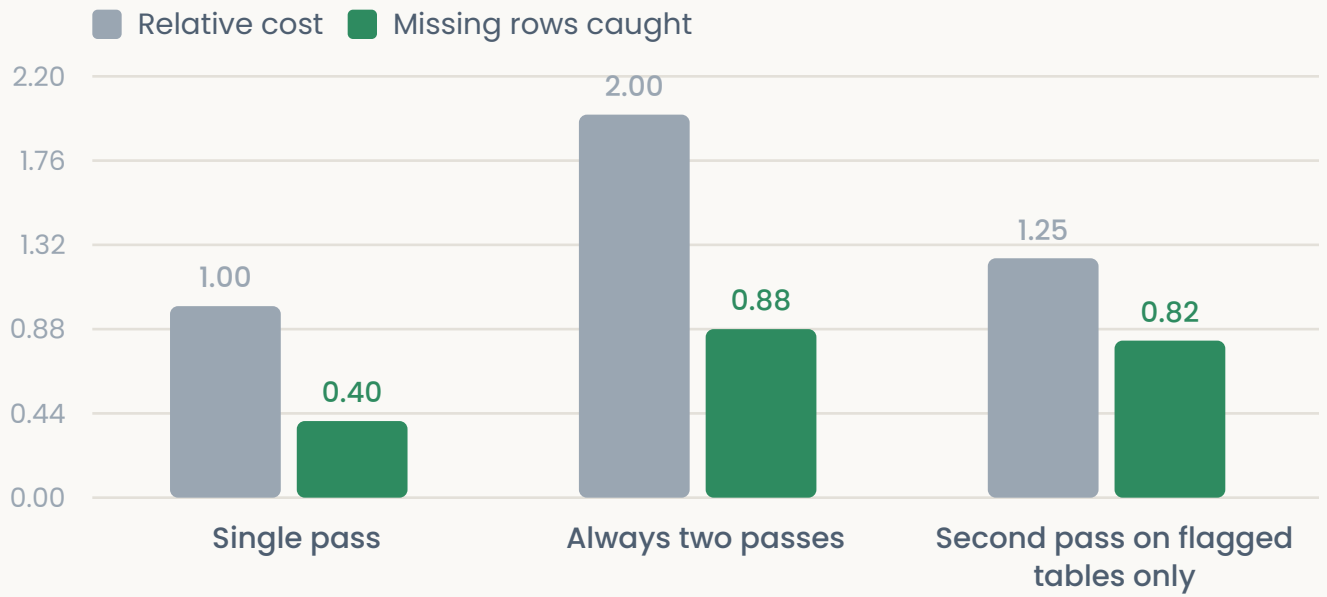
Missed

20.4 - 20 = **≈0.4 rows** per table. Across 500 tables, about 200 rows never extracted by anything.

Assumption: the passes must be independent — two prompts on one model miss the same rows.

THE ECONOMICS

Spend the Second Pass Selectively



Layer 1 does the triage.

Deterministic checks are nearly free and flag the risky tables. The expensive second pass runs only on those.

Most of the benefit, a quarter of the cost.

Illustrative ratios.

RECAP

What We Measure From Monday

1. **Row recall, bucketed by table length**
   Not document accuracy. Not field accuracy.

2. **Flag rate from structural checks**
   And what share of flags are real misses.

3. **Estimated misses per table**
   Capture–recapture on a sampled subset.

A wrong value has an advocate. A missing row does not. **That is the whole argument.**